

MSGF V3.2: ULTRA-PERFORMANCE SPEC

ELPHIE SYNTAX LLC - STATEFUL LOGIC ARCHITECTURE

I. HIGH-SPEED STORAGE & SHARDING

Hot Layer (Redis)

In-memory data store for the **P4 State Ledger**. Handles nanosecond-speed "Active Slices" of the current logic gate to eliminate system lag.

Cold Layer (Postgres)

Long-term archival of the **6-Pillar Core**. Uses `pgvector` for semantic retrieval of historical data and the 1.1.1 Index.

II. THE 1.1.1 GENEALOGICAL BUG INDEX

Bugs are sharded into a hierarchical tree to enable the AI to perform "lineage scans" and cross-reference fixes efficiently.

- 1.0 Category:** Major module (e.g., Auth, UI, API).
- 1.1 Branch:** Specific sub-logic or LOM gate.
- 1.1.1 Instance:** The discrete "Fix Delta" and specific P4 snapshot.

III. THE VAULT vs. THE HALL OF HALLUCINATIONS

Differential state management ensures the system only learns from meaningful data.

The Vault (Positive Index)	The Hall of Hallucinations (Negative Index)
Stores successful 1.1.1 fixes. The AI cross-references this to repeat "What Worked."	Stores failed Shadow Mode attempts and rejected consensus logic. The AI cross-references this to avoid "What Failed."

**Automatic Purge: Hall of Hallucinations is cleared of non-critical "Low" tier data every 30 days to optimize vector search speed.*

IV. TIERED BATCHING & HUMAN TIE-BREAKER

Tier	Frequency	Action Protocol
RED (Critical)	Immediate	Claude/Gemini Consensus + Shadow Mode. Human Tie-Breaker mandatory on disagreement.
YELLOW (Med)	6 Hours	Batched summary for periodic calibration.
GREEN (Low)	24 Hours	Cumulative style and token efficiency reports.

V. MASTER DIRECTIVE (V3.2-ULTRA)

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INIT MSGF V3.2-ULTRA:
1. SWEEP: Perform Day-Zero scan; export pre_ingestion_audit.md.
2. SHARD: Load Active Gates into Redis (Hot); map 6-Pillars to Postgres (Cold).
3. DEFEND: Enable Shadow Mode and LOM Gates for all operations.
4. CROSS-REF: Pre-flight check comparing proposed fixes against THE VAULT and THE HALL.
5. CONVERGE: Execute Claude/Gemini Consensus.
6. ARBITRATE: Halt and trigger Human Tie-Breaker if Consensus fails or Retry > 3.
7. PERSIST: Save successful Deltas to The Vault; discard redundant State data.
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